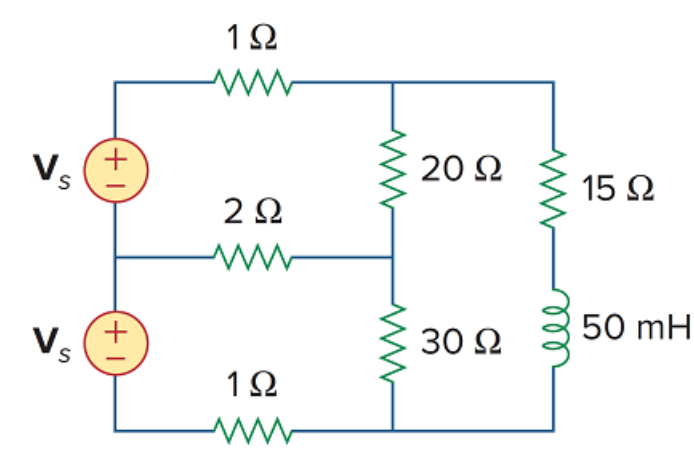


Problem

Consider the single-phase three-wire system shown in Fig. 12.78. Find the current in the neutral wire and the complex power supplied by each source. Take $\mathbf{V}_s$ as a $220\angle 0^\circ$ -V, 60-Hz source.

Figure 12.78



Step-by-step solution

Step 1 of 9

Refer to Figure 12.78 in the text book.

Calculate the value of the reactance due to inductance.

$$X_L = 2\pi fL$$

Substitute 60 Hz for f and 50 mH for L .

$$\begin{aligned} X_L &= j2\pi(60)(50 \text{ m}) \\ &= j18.85 \Omega \end{aligned}$$

Therefore, the value of the reactance X_L is, $j18.85 \Omega$.

Step 2 of 9

Identify the loops and currents in the circuit diagram shown in Figure 1.

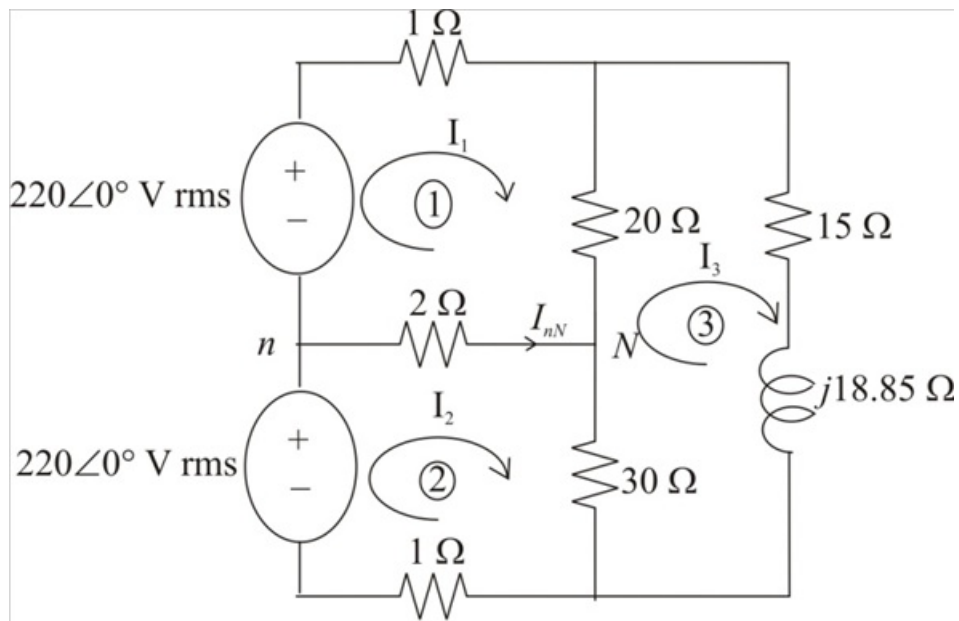


Figure 1

Step 3 of 9

Apply Kirchhoff's Voltage Law in loop-3.

$$I_3(15 + j18.85) + (I_3 - I_2)(30) + (I_3 - I_1)(20) = 0$$

$$I_3(15 + j18.85 + 20 + 30) - 30I_2 - 20I_1 = 0$$

$$I_3(65 + j18.85) - 30I_2 - 20I_1 = 0$$

$$I_3(65 + j18.85) = 30I_2 + 20I_1$$

$$I_3 = \frac{30I_2 + 20I_1}{65 + j18.85}$$

Therefore, the expression for the current I_3 is, $\frac{30I_2 + 20I_1}{65 + j18.85}$.

Step 4 of 9

Apply Kirchhoff's Voltage Law in loop-2.

$$-220\angle 0^\circ + (I_2 - I_1)(2) + (I_2 - I_3)(30) + I_2(1) = 0$$

$$I_3(-30) + I_2(2 + 30 + 1) - 2I_1 - 220\angle 0^\circ = 0$$

$$-30I_3 + 33I_2 - 2I_1 - 220\angle 0^\circ = 0$$

Substitute $\frac{30I_2 + 20I_1}{65 + j18.85}$ for I_3 .

$$-30\left(\frac{30I_2 + 20I_1}{65 + j18.85}\right) + 33I_2 - 2I_1 - 220\angle 0^\circ = 0$$

$$\frac{-900I_2 - 600I_1}{65 + j18.85} + 33I_2 - 2I_1 - 220\angle 0^\circ = 0$$

$$I_2\left(\frac{-900}{65 + j18.85} + 33\right) - I_1\left(\frac{600}{65 + j18.85} + 2\right) - 220\angle 0^\circ = 0$$

$$I_2(13.3\angle 163.83^\circ + 33) - I_1(8.87\angle -16.17^\circ + 2) - 220\angle 0^\circ = 0$$

$$I_2(20.56\angle 10.38^\circ) - I_1(10.81\angle -13.22^\circ) - 220\angle 0^\circ = 0$$

$$I_2(20.56\angle 10.38^\circ) = I_1(10.81\angle -13.22^\circ) + 220\angle 0^\circ$$

$$I_2 = \frac{I_1(10.81\angle -13.22^\circ) + 220\angle 0^\circ}{20.56\angle 10.38^\circ} \dots\dots (1)$$

Therefore, the expression for the current I_2 is, $\frac{I_1(10.81\angle -13.22^\circ) + 220\angle 0^\circ}{20.56\angle 10.38^\circ}$.

Step 5 of 9

Apply Kirchhoff's Voltage Law in loop-1.

$$-220\angle 0^\circ + (I_1 - I_2)(2) + (I_1 - I_3)(20) + I_1(1) = 0$$

$$I_3(-20) - I_2(2) + I_1(2 + 20 + 1) - 220\angle 0^\circ = 0$$

$$-20I_3 - 2I_2 + 23I_1 - 220\angle 0^\circ = 0$$

Substitute $\frac{30I_2 + 20I_1}{65 + j18.85}$ for I_3 .

$$-20\left(\frac{30I_2 + 20I_1}{65 + j18.85}\right) - 2I_2 + 23I_1 - 220\angle 0^\circ = 0$$

$$\frac{-600I_2 - 400I_1}{65 + j18.85} - 2I_2 + 23I_1 - 220\angle 0^\circ = 0$$

$$I_2\left(\frac{-600}{65 + j18.85} - 2\right) + I_1\left(23 - \frac{400}{65 + j18.85}\right) - 220\angle 0^\circ = 0$$

$$I_2(-8.87\angle 163.83^\circ - 2) + I_1(23 - 5.91\angle -16.18^\circ) - 220\angle 0^\circ = 0$$

$$I_2(10.81\angle 166.79^\circ) + I_1(17.4\angle 5.43^\circ) - 220\angle 0^\circ = 0$$

Substitute $\frac{I_1(10.81\angle -13.22^\circ) + 220\angle 0^\circ}{20.56\angle 10.38^\circ}$ for I_2 .

$$\left(\frac{I_1(116.86\angle 153.57^\circ) + 2378.2\angle 166.79^\circ}{20.56\angle 10.38^\circ}\right) + I_1(17.4\angle 5.43^\circ) - 220\angle 0^\circ = 0$$

$$I_1(5.68\angle 143.19^\circ) + 115.67\angle 156.41^\circ + I_1(17.4\angle 5.43^\circ) - 220\angle 0^\circ = 0$$

$$I_1(13.74\angle 21.57^\circ) + 329.27\angle -171.92^\circ = 0$$

$$I_1(13.74\angle 21.57^\circ) = -329.27\angle -171.92^\circ$$

$$I_1(13.74\angle 21.57^\circ) = 329.27\angle 8.08^\circ$$

$$I_1 = \frac{329.27\angle 8.08^\circ}{13.74\angle 21.57^\circ}$$

$$I_1 = 23.96\angle -13.49^\circ \text{ A}$$

Therefore, the value of the current I_1 is $23.96\angle -13.49^\circ \text{ A}$.

Step 6 of 9

Recall equation (1).

$$I_2 = \frac{I_1(10.81\angle -13.22^\circ) + 220\angle 0^\circ}{20.56\angle 10.38^\circ}$$

Substitute $23.96\angle -13.49^\circ$ A for I_1 .

$$\begin{aligned} I_2 &= \frac{(23.96\angle -13.49^\circ)(10.81\angle -13.22^\circ) + 220\angle 0^\circ}{20.56\angle 10.38^\circ} \\ &= \frac{259\angle -26.71^\circ + 220\angle 0^\circ}{20.56\angle 10.38^\circ} \\ &= \frac{466.13\angle -14.46^\circ}{20.56\angle 10.38^\circ} \\ &= 22.67\angle -24.84^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current I_2 is, $22.67\angle -24.84^\circ$ A.

Step 7 of 9

Calculate the value of the current I_{nN} .

$$I_{nN} = I_2 - I_1$$

Substitute $22.67\angle -24.84^\circ$ A for I_2 and $23.96\angle -13.49^\circ$ A for I_1 .

$$\begin{aligned} I_{nN} &= 22.67\angle -24.84^\circ - 23.96\angle -13.49^\circ \\ &= 4.79\angle -124.72^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current I_{nN} is, $4.79\angle -124.72^\circ$ A.

Step 8 of 9

Calculate the complex power supplied by the source in loop-1.

$$S_1 = (220\angle 0^\circ)I_1^*$$

Substitute $23.96\angle -13.49^\circ$ A for I_1 .

$$\begin{aligned} S_1 &= (220\angle 0^\circ)(23.96\angle -13.49^\circ)^* \\ &= (220\angle 0^\circ)(23.96\angle 13.49^\circ) \\ &= 5.13 + j1.23 \text{ kVA} \end{aligned}$$

Therefore, the value of the complex power S_1 is, $5.13 + j1.23$ kVA.

Step 9 of 9

Calculate the complex power supplied by the source in loop-2.

$$S_2 = (220\angle 0^\circ)I_2^*$$

Substitute $22.67\angle -24.84^\circ$ A for I_2 .

$$\begin{aligned} S_2 &= (220\angle 0^\circ)(22.67\angle -24.84^\circ)^* \\ &= (220\angle 0^\circ)(22.67\angle 24.84^\circ) \\ &= 4.53 + j2.1 \text{ kVA} \end{aligned}$$

Therefore, the value of the complex power S_2 is, $4.53 + j2.1$ kVA.